INTRODUCTION TO C

C is a

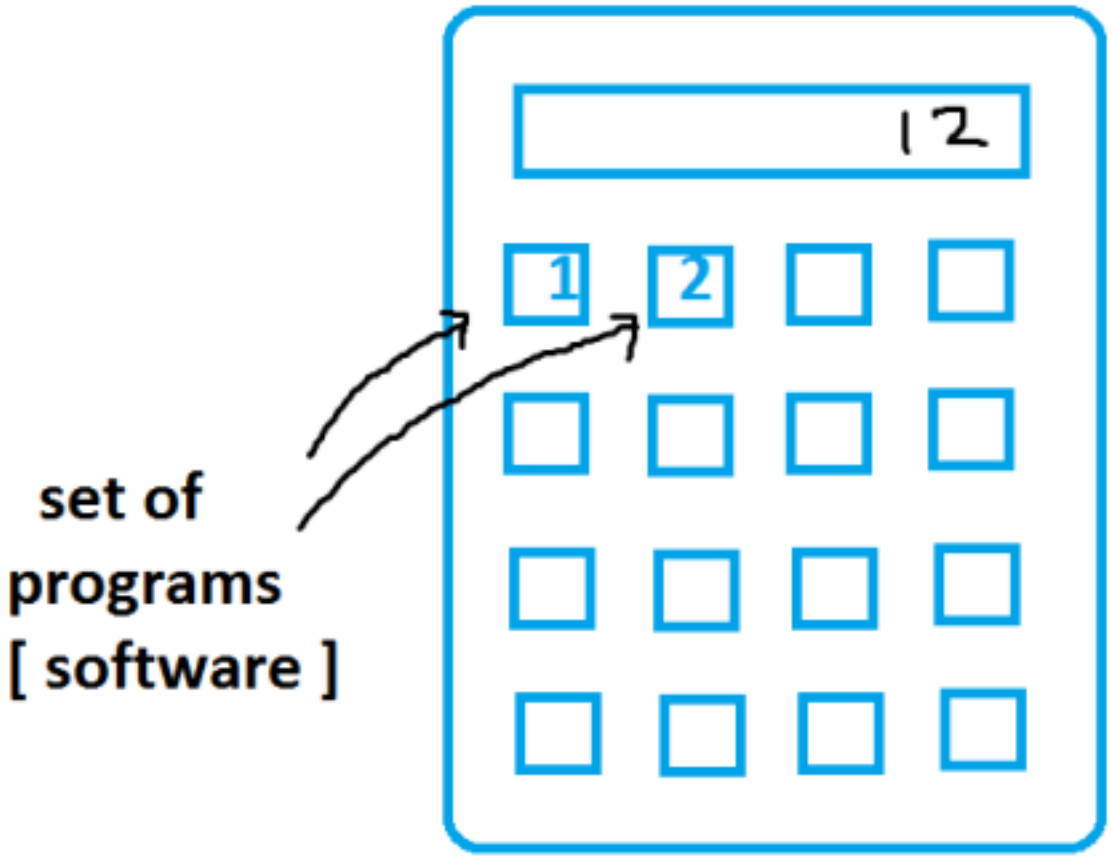
1. **It is a high level / middle level programming language.**
2. **It is a compiler based programming language.** 3. **It is a procedure oriented programming language [POP]**
4. **It is a general purpose programming language.**

What is a program?

Set of instructions is called program.

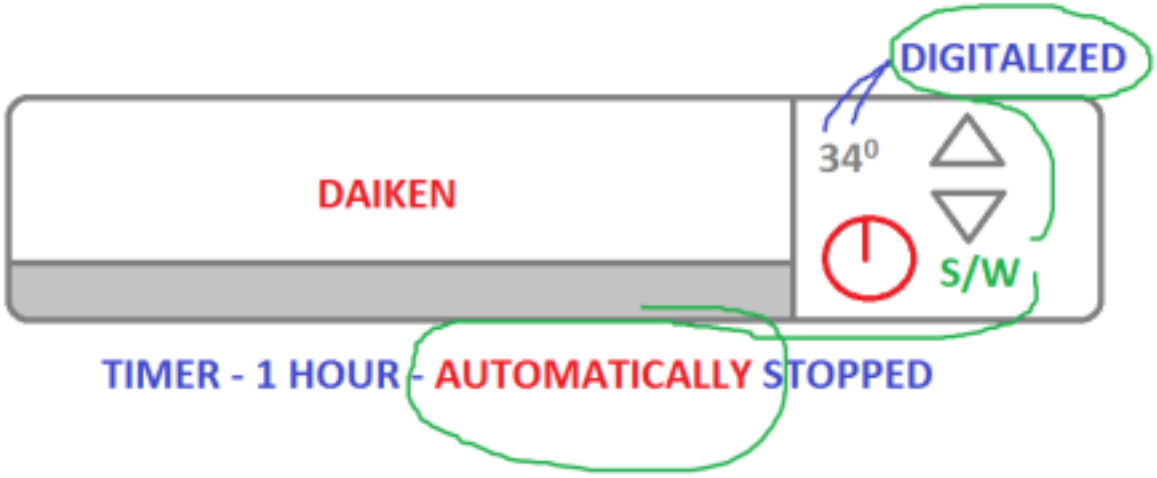
What is a software?

Set of programs is called software.



Or

As per IT Industry software is a digitalized and automated process.



Basically the software divided into 2 types.

1. **System software**

Eg. o.s , device drivers, translators

2. Application software

   Eg: whatsapp, phonepe,…

What is a language?

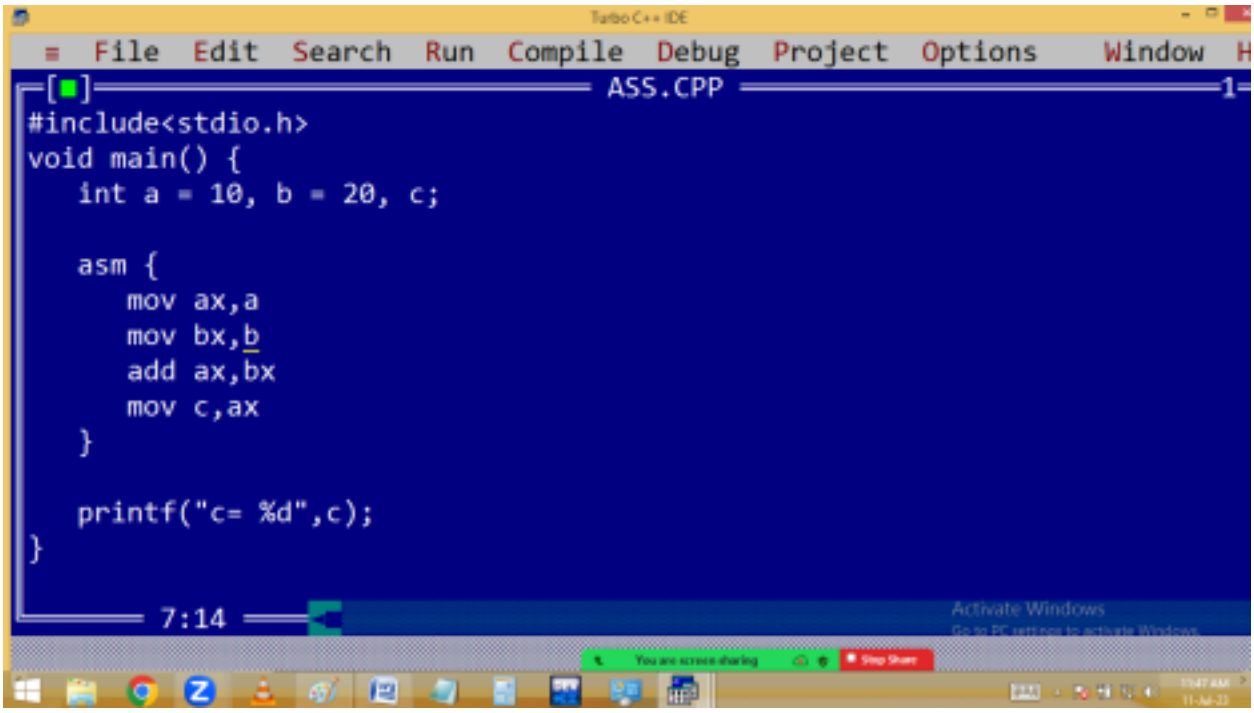
Generally the languages like telugu, English, hindi etc are used to communicate with humans. Hence they are called human languages.

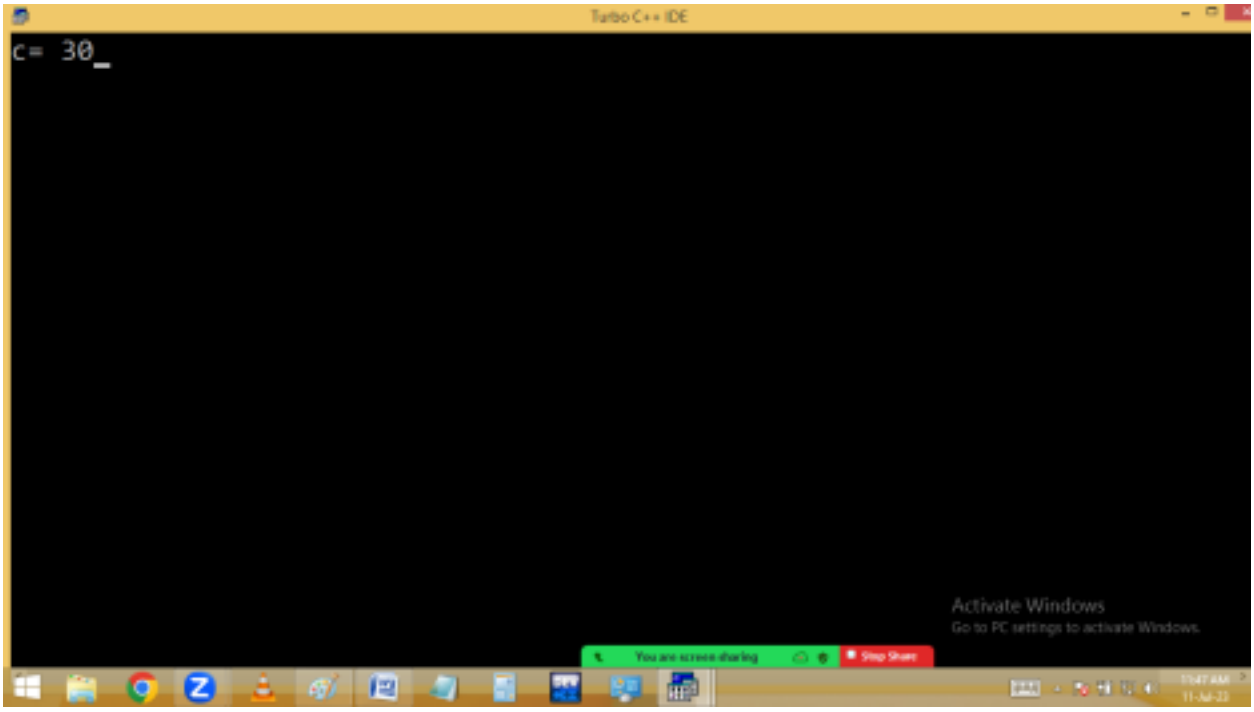
To communicate with machines we are using the computer programming languages like C / C++ / Java / .Net / py etc. i.e. they are used to write the programs [software] to communicate with the machines.

Basically the computer languages divided into 3 types.

1. Machine language: Created with binary code. Eg: 10101110
2. Low level / assembly languages: Created with English like shortcuts called MNEMONICS. Eg: sub, add

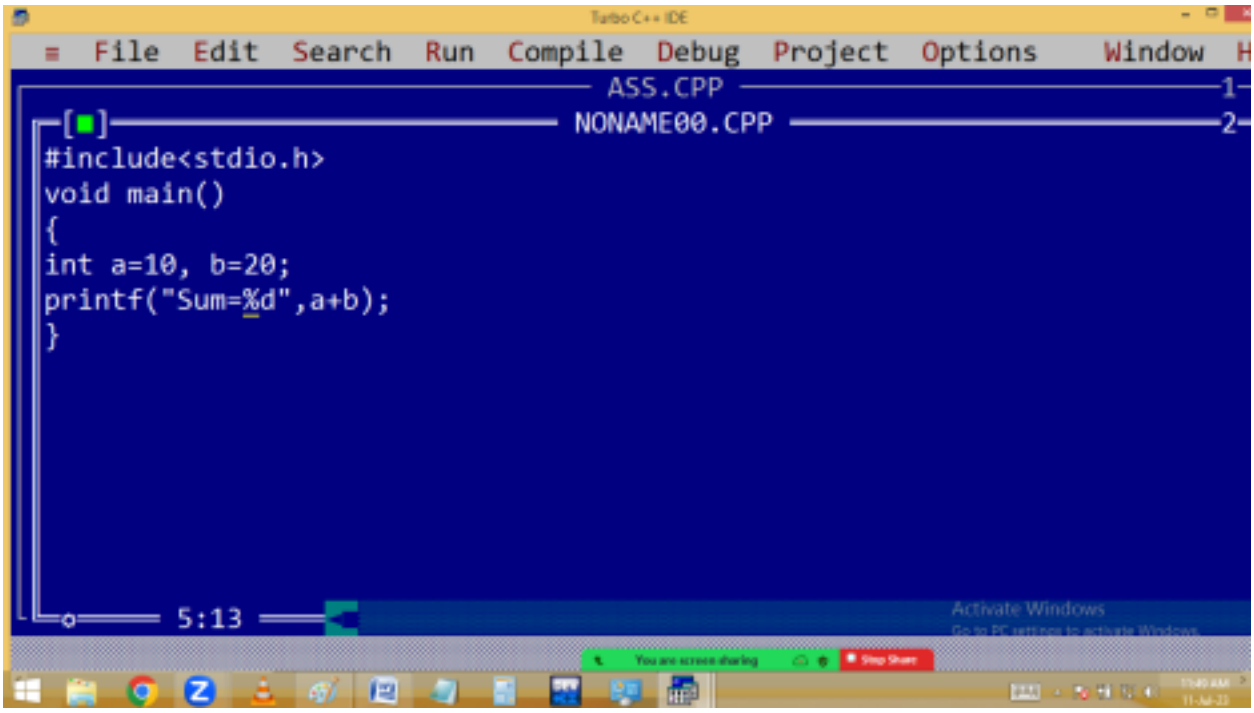
   Example:





3.High level languages: created with simple English.

Eg: subject, addition,..



C comes with both low level and high level features. Hence c is a middle level language.

Because of the middle level features, we can develop both system software [low level features] and application software [high level features]. Hence c is a multipurpose

programming language.

What is a translator?

Always the user given instructions are in English, which is called source code or source program. But the computer understandable code is binary code / machine language. to convert this source code into binary code and to check the errors we are using the translators like

1.Compiler

2.Interpreter

3.Assembler

Compiler and interpreter both used to convert

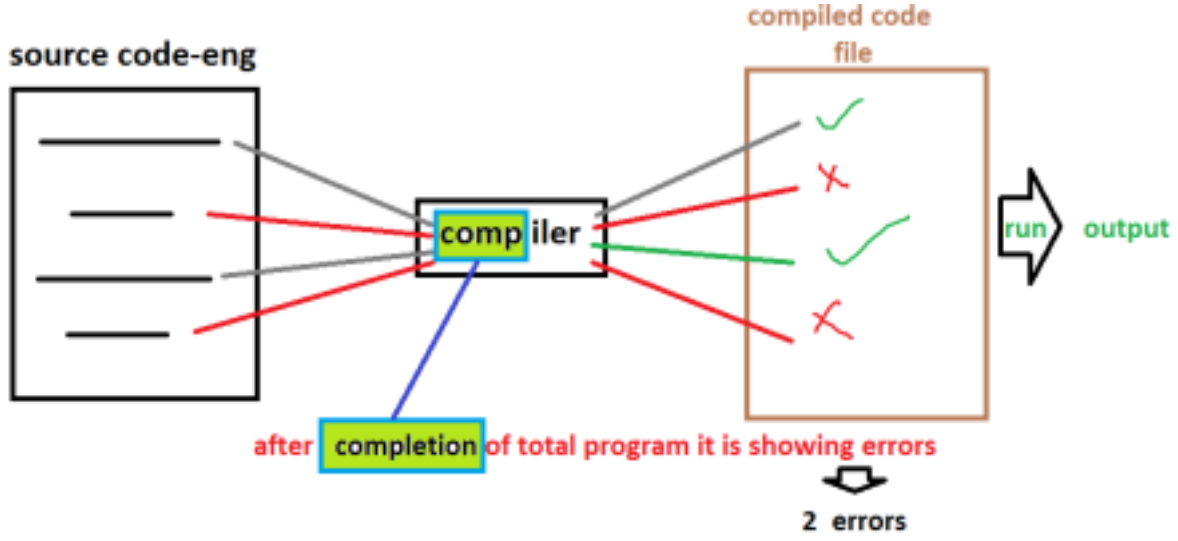
high level programs to machine language.

Assembler is used to convert low level programs to machine language.

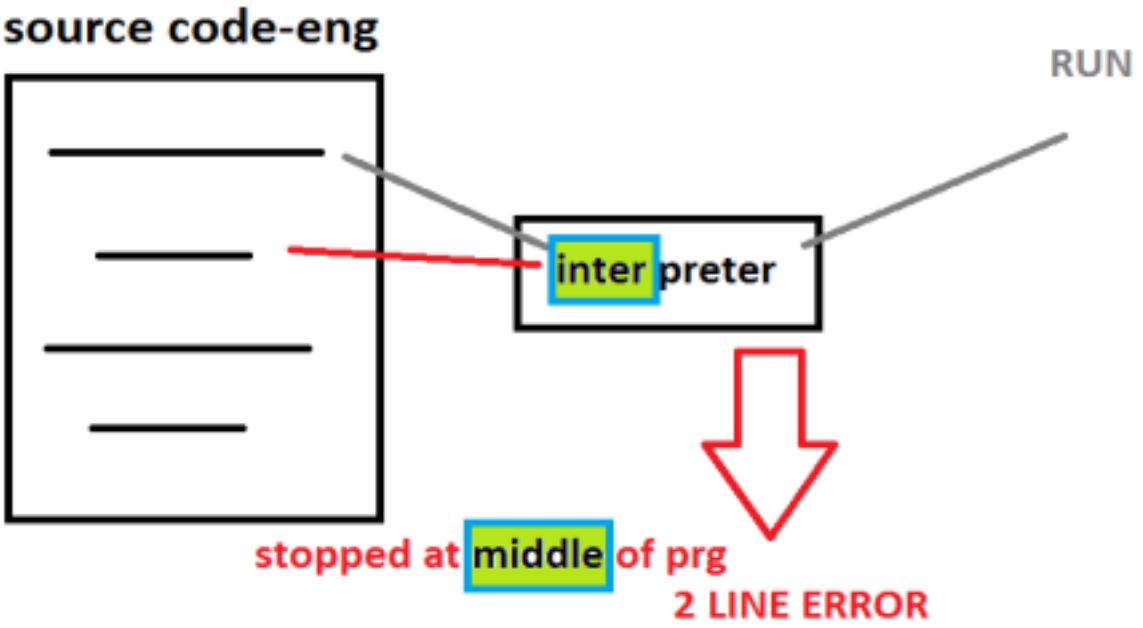
```
source code / source program
        english
#include<stdio.h>
void main()
{
int a=10, b=20;
printf("Sum=%d",a+b);
}
```

Translators
1. source code to machine language/binary code
2. error checking

Compiler shows the errors after completion of total source code. i.e. compiler completes the total source code at once by leaving error lines.



Interpreter checks line by line. i.e. when any error is occurred, stop the process until that error is solved.



Assembler working style is similar to compiler.
C & C++ are using compilers. Hence they are called compiler based programming languages. java / .net / py are using both compilers and interpreters. Hence they are called compiler based interpreted languages.

What is programming paradigm?

Every programming language is having particular structure with rules and regulations, which is technically called programming paradigm.

Before c language the programming languages are using monolithic programming.

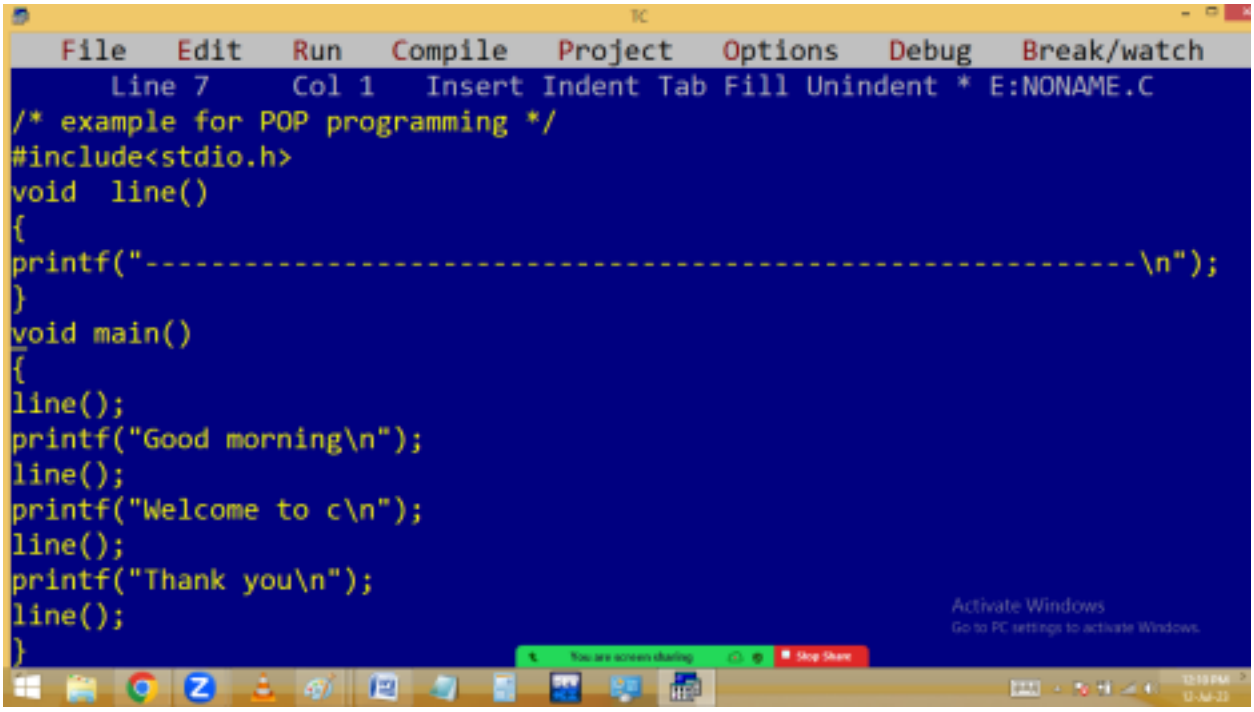
```
File   Edit   Run   Compile   Project   Options   Debug   Break/watch
     Line 1     Col 41  Insert Indent Tab Fill Unindent * E:NONAME.C
/* example for monolithic programming */_
#include<stdio.h>
void  main()
{
printf("------------------------------------------------------------\n");
printf("Good morning\n");
printf("------------------------------------------------------------\n");
printf("Welcome to c\n");
printf("------------------------------------------------------------\n");
printf("Thank you\n");
printf("------------------------------------------------------------");
}
```

```
------------------------------------------------------------
Good morning
------------------------------------------------------------
Welcome to c
------------------------------------------------------------
Thank you
------------------------------------------------------------_
```

In monolithic programming we are creating the entire program using single program. Due to this it is very difficult to find the errors, takes more memory and performance is poor.

To avoid this program they have introduced

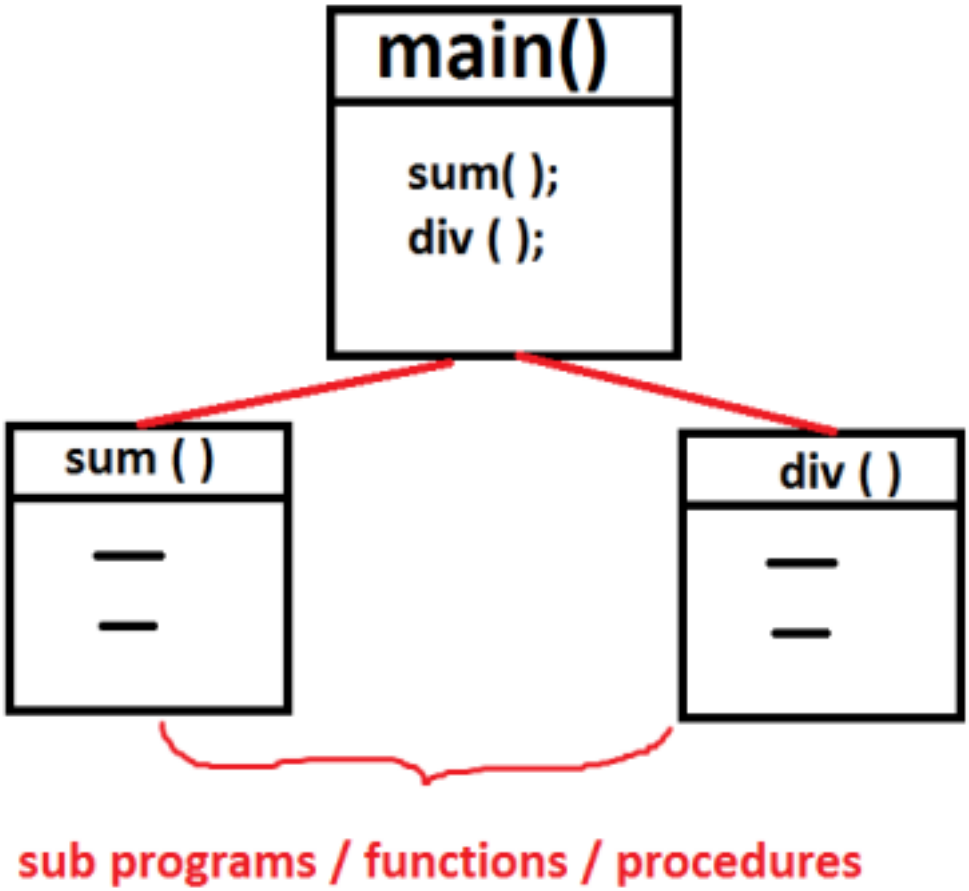
procedure oriented programming. In pop a big program divided into several small sub program / sub routines / procedures / functions / modules / structures/. Hence c program is collections of procedures, it is called procedure oriented programming.



Advantages:

1. **Modularity: Dividing a big program into several small modules.**
2. **Reusability: Write once, use many times.**
3. **Simplicity: Easy to understand / read.**
4. **Efficiency: performance is high**

Procedure oriented structure:


main()
sum( );
div ( );
sum ( )
div ( )
sub programs / functions / procedures